3.5 Workflow Optimisation Options

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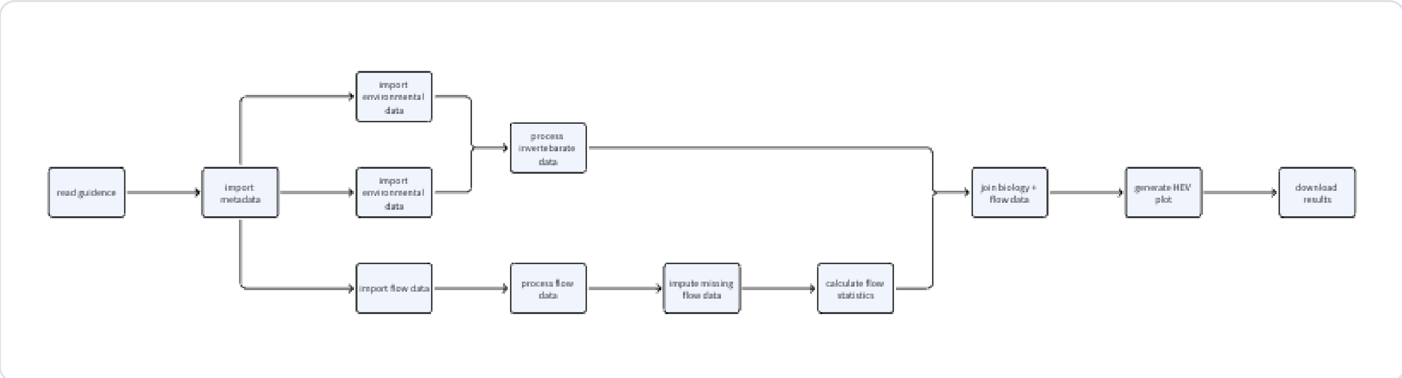
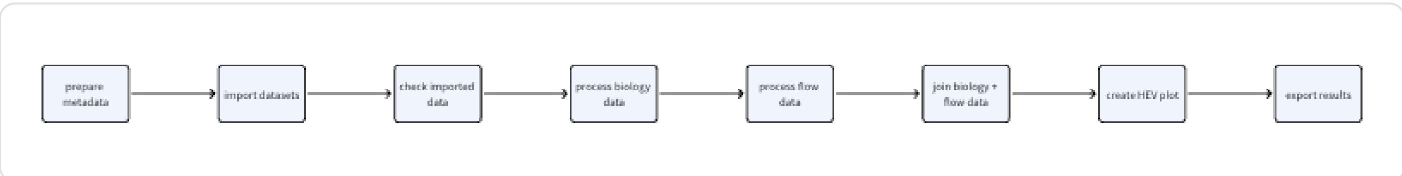
The current HE Toolkit Dashboard already supports the main analytical workflow, from importing metadata and datasets to processing biology and flow data, joining hydroecological data, and generating HEV plots.



However, the workflow can still be difficult for first-time non-coder users because the dependencies between steps are not always obvious. Three possible workflow optimisation directions are proposed below.

Option 1: Step-by-Step Guided Workflow

This option restructures the dashboard into a clear linear workflow:



In this design, users would move through the dashboard step by step. Each step would show the current task, required inputs, expected outputs, and a clear “Next” action. Later steps would remain disabled until the required previous steps are completed.

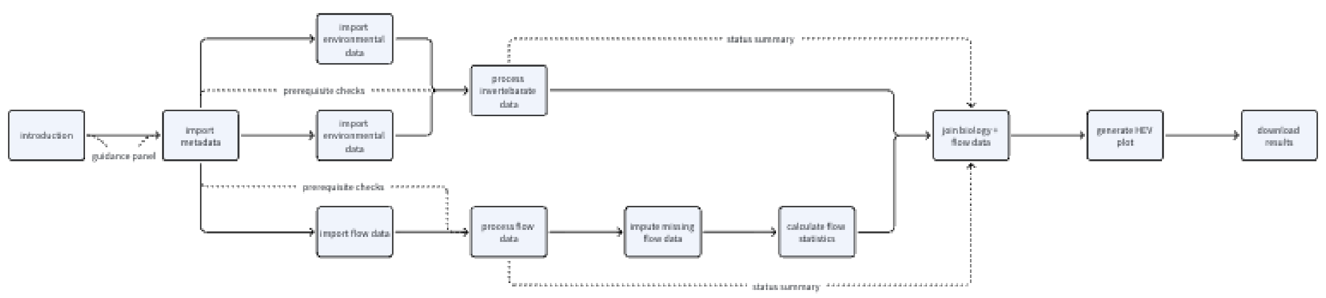
This approach would help non-coder users understand where they are in the process, what they need to do next, and why each step is necessary. It would also reduce the chance of users clicking processing buttons before the required data are available.

The main advantage of this option is that it provides the strongest guidance for first-time users. The main disadvantage is that it would require a larger redesign of the current interface structure.

Option 2: Add Workflow Guidance to the Existing Dashboard

This option keeps the current page structure but adds clearer workflow guidance, validation and status information to each page.

The dashboard would still contain the existing pages:



However, each page would include a workflow status panel showing the current step, completed steps, required prerequisites, and the recommended next step. Processing buttons could be disabled until the required input data are available. Tooltips or short help text could also be added for technical terms such as RICT predictions, O:E ratios, donor flow sites, lag, join method and flow statistics.

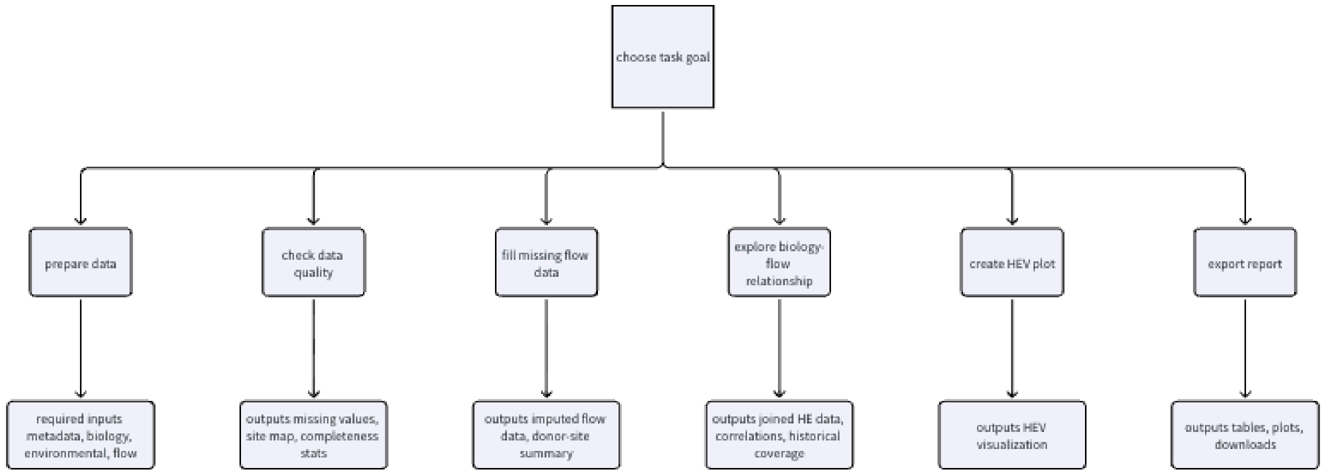
This option would improve usability without completely rebuilding the dashboard. It is a practical short-term improvement because it builds on the existing structure.

The main advantage is that it is easier to implement. The main disadvantage is that the dashboard would still be organised around technical modules, so some users may still need extra guidance to understand the full workflow.

Option 3: Task-Oriented Workflow

This option reorganises the dashboard around user goals.

Instead of asking users to follow functions such as importing data, running RICT predictions, calculating flow statistics and joining data, the dashboard would present task-based options such as:



In this design, users would start by selecting what they want to achieve. The dashboard would then guide them through the required inputs and background processing. Advanced parameters could be hidden by default, while the interface focuses on meaningful tasks and outputs.

This approach is closest to a decision-support tool. It would be especially useful for Environment Agency staff or other users who are more interested in interpreting results than understanding the underlying R workflow.

The main advantage is that it best matches real user goals. The main disadvantage is that it would require the largest redesign, because the current technical workflow would need to be reorganised behind a more user-centred interface.